

Macro-Aware Portfolio Risk Analysis

Explained project portfolio - Python, FactSet data and macro-factor regression

Project objective

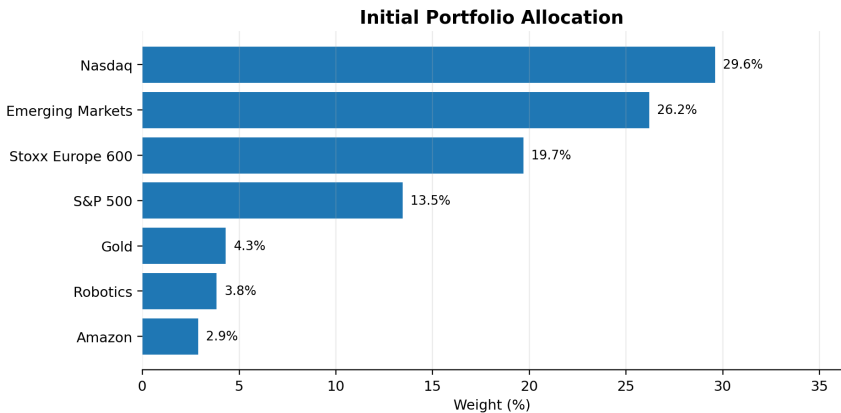
This project analyzes a real personal multi-asset portfolio built before a more advanced finance toolkit was developed. The goal was not simply to report performance, but to understand whether the allocation was genuinely diversified, which risks dominated the portfolio, how macroeconomic shocks affected returns, and how a constrained Markowitz + macro allocation framework could improve the decision process.

What the analysis actually did

- 1 Built the weighted portfolio return series from FactSet asset returns and actual invested amounts.
- 2 Measured historical performance, drawdowns, monthly VaR and CVaR at portfolio and asset level.
- 3 Created a monthly macro dataset with rates, inflation, VIX, yield curve and dollar variables.
- 4 Ran OLS regressions with HAC robust errors to estimate macro sensitivity and standardized beta exposure.
- 5 Implemented a constrained grid-search allocation combining Markowitz metrics and macro-risk penalties.

Input data and return construction

The starting point was the real portfolio allocation. Each asset return series was imported from FactSet files, cleaned, converted to numeric values, aligned on common dates and converted into decimal returns when necessary. The portfolio return was then computed as a weighted sum of the asset returns using the actual investment weights.



Core formula

Portfolio return: $R_{p,t} = \sum_i w_i \times R_{i,t}$. Monthly returns were obtained by compounding daily returns: $R_m = \text{Product}(1 + R_d) - 1$. This makes the analysis consistent across daily, monthly and yearly frequencies.

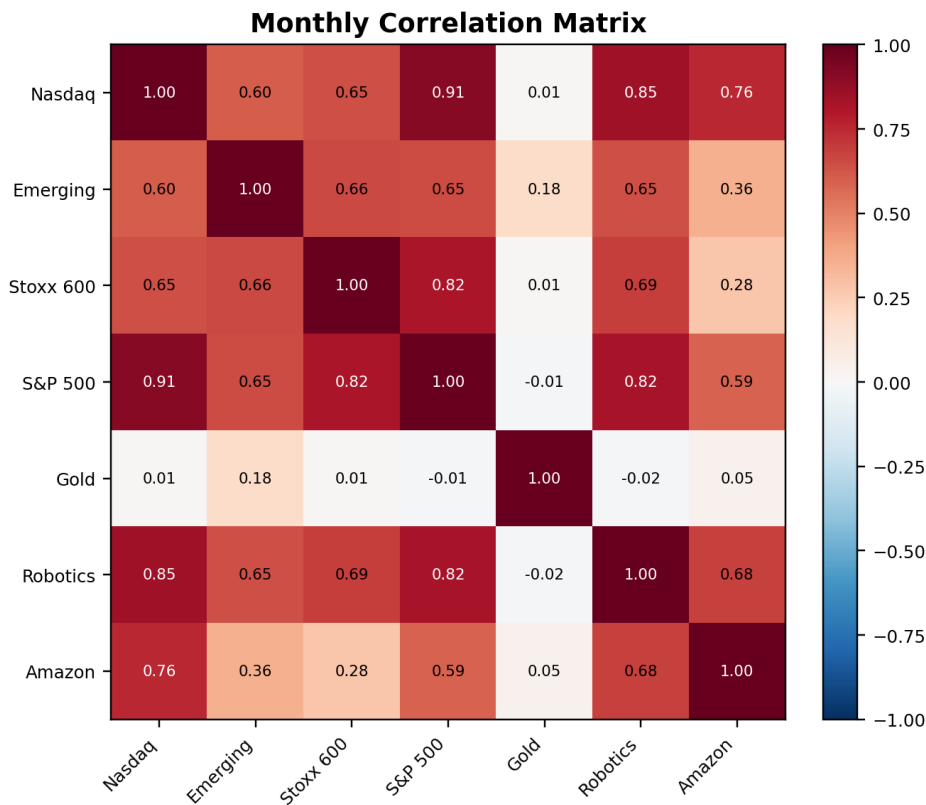
Risk and performance modules

Module	What it measures	Why it matters
Performance	Capital curve, return, Sharpe, Sortino	Return quality, not only final profit
Downside risk	Drawdown, VaR, CVaR	Loss severity and tail events
Diversification	Correlation matrix	Hidden common risk factors
Macro exposure	OLS regression and beta	Sensitivity to rates, VIX, inflation
Allocation	Markowitz + macro score	Improvement scenario under constraints

Asset	Initial weight
Nasdaq	29.62%
Emerging Markets	26.20%
Stoxx Europe 600	19.70%
S&P 500	13.46%
Gold	4.31%
Robotics	3.84%
Amazon	2.88%

Initial diagnosis

The allocation looked diversified by number of instruments, but the analytical question was whether it was economically diversified after accounting for correlations, downside risk and macro exposure.



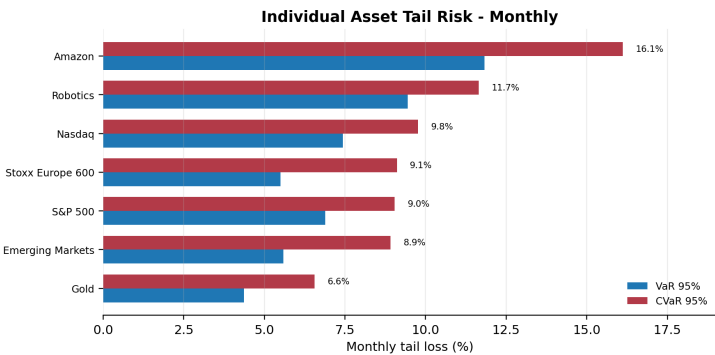
What the correlation matrix shows

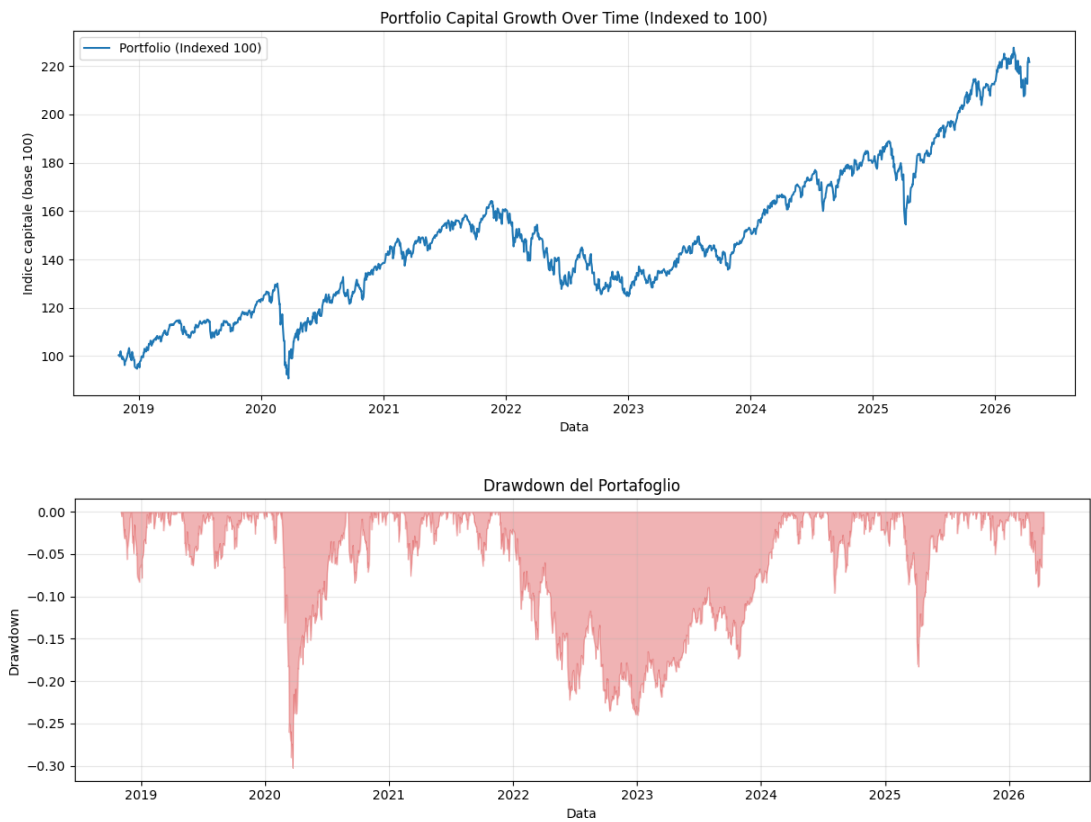
The portfolio was not equally diversified across independent risk sources. Nasdaq, S&P; 500, Robotics and Amazon show strong positive correlations, which means that several positions were likely to fall together during equity/growth sell-offs. Gold was the main low-correlation exposure: it had near-zero monthly correlation with Nasdaq and S&P; 500.

Relationship	Monthly corr.	Meaning
Nasdaq vs S&P 500	0.91	Very similar equity beta
Nasdaq vs Robotics	0.85	Growth/theme risk overlap
S&P 500 vs Robotics	0.82	High shared market factor
Gold vs S&P 500	-0.01	Potential diversifier
Gold vs Nasdaq	0.01	Low common movement

Which assets carried the largest tail risk?

To identify which positions were most fragile in downside months, individual monthly VaR and CVaR were computed. This complements the macro regression: the regression explains portfolio sensitivity to macro factors, while VaR/CVaR shows which assets had the largest standalone tail losses.





How VaR and CVaR were used

Historical VaR at 95% estimates the loss threshold exceeded in the worst 5% of observations. CVaR at 95% goes further: it measures the average loss inside that worst 5% tail. Therefore, CVaR is more useful when the objective is to understand how severe losses can be when the VaR threshold is breached.

Risk metric	Daily	Monthly	Yearly
VaR 95%	1.60%	6.79%	14.69%
CVaR 95%	2.57%	7.68%	22.11%
VaR 99%	3.19%	7.80%	20.62%
CVaR 99%	4.17%	9.00%	22.11%

Historical growth

>2.2x

indexed base 100

Worst drawdown

~ -30%

COVID shock

Monthly VaR 95%

6.79%

tail threshold

Monthly CVaR 95%

7.68%

avg. tail loss

Monthly obs.

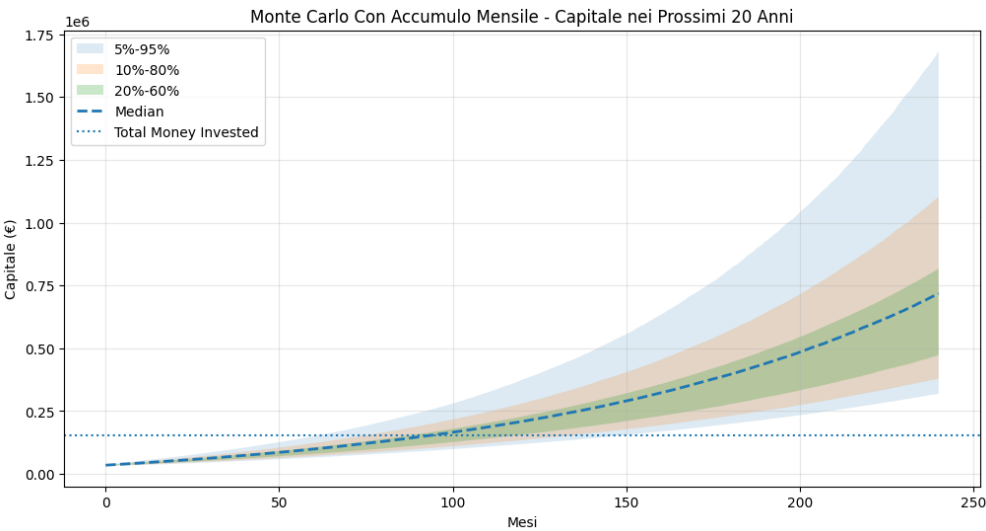
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2018-2026 sample

Model setup

The Monte Carlo engine used a bootstrap approach: each simulated month randomly samples one historical monthly portfolio return, with replacement. The 20-year simulation starts from EUR 34,776 and adds EUR 500 at the end of each month after applying the sampled return. This produces a distribution of possible capital paths rather than a single deterministic forecast.

Input / output	Value
Number of simulations	50,000
Investment horizon	20 years / 240 months
Initial capital	EUR 34,776
Monthly contribution	EUR 500
Total money invested	EUR 154,776
Median final capital	EUR 718,144
Mean final capital	EUR 822,580
5th percentile	EUR 317,709
95th percentile	EUR 1,682,774



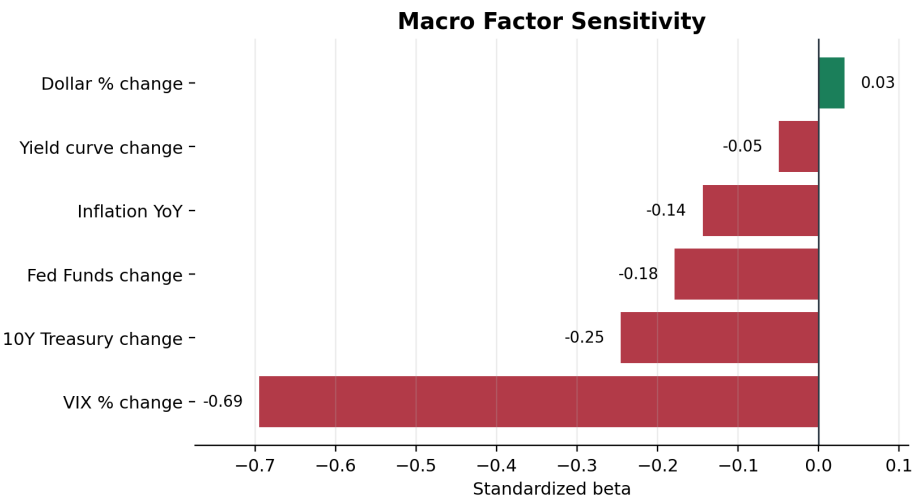
Interpretation discipline

The fan chart is useful for long-term planning because it shows dispersion and downside scenarios, not only the median outcome. However, the model assumes the historical monthly return distribution remains informative. It does not include taxes, transaction costs, regime changes, personal liquidity needs or future structural breaks.

How macro variables were created

The analysis merged monthly portfolio returns with macro variables representing rates, inflation, risk sentiment and the dollar. Raw series included Fed Funds Rate, 2Y and 10Y Treasury yields, CPI, VIX and a broad dollar index. The model then created changes and percentage changes to capture shocks rather than only levels.

Variable	Transformation	Interpretation
DGS10_Change	monthly diff.	10Y yield shock
DFF_Change	monthly diff.	Fed policy shock
Curve_Change	diff. of 10Y-2Y	yield-curve shock
Inflation_YoY	CPI pct change 12m	inflation regime
VIX_Pct_Change	VIX pct change	risk sentiment shock
Dollar_Pct_Change	dollar pct change	USD strength shock



Regression model

Monthly portfolio return was regressed on the macro factors: $R_{p,t} = \alpha + \beta_1 \text{DGS10_Change} + \beta_2 \text{DFF_Change} + \beta_3 \text{Curve_Change} + \beta_4 \text{Inflation_YoY} + \beta_5 \text{VIX_Pct_Change} + \beta_6 \text{Dollar_Pct_Change} + \text{error}$. HAC robust standard errors with 3 lags were used to reduce time-series inference bias.

Metric	Value
Observations	77 months
R-squared	0.4798
Adjusted R-squared	0.4352
Mean actual return	0.87% monthly

Why standardized betas matter

Raw coefficients use different units. Standardized betas put all variables on a comparable scale, making it easier to see that VIX shocks dominated the macro sensitivity of the portfolio.

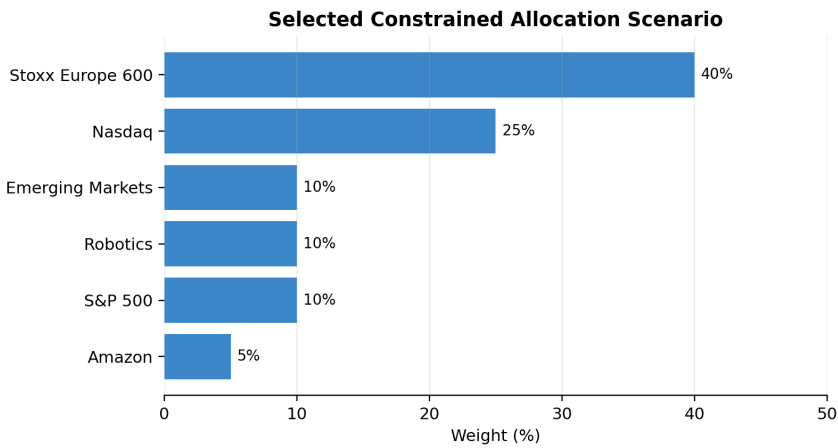
Main macro-regression results

Factor	Raw coeff.	Std beta	t-stat	p-value	Sig.
VIX % change	-0.1018	-0.6948	-8.337	0.0000	***
10Y Treasury change	-0.0452	-0.2459	-2.247	0.0246	**
Fed Funds change	-0.0324	-0.1790	-1.217	0.2238	
Inflation YoY	-0.2549	-0.1441	-0.954	0.3403	
Yield curve change	-0.0146	-0.0494	-0.517	0.6055	
Dollar % change	0.1028	0.0321	0.244	0.8070	

The result is economically intuitive for a growth-oriented equity portfolio. The strongest negative standardized beta is linked to VIX percentage changes, meaning the portfolio is highly exposed to volatility/risk-off shocks. The 10Y Treasury change is also negative and statistically significant, consistent with sensitivity to discount-rate repricing. Fed Funds, inflation, curve and dollar variables are less statistically strong in this specific sample.

How Markowitz was implemented

The allocation exercise used a grid search, not a black-box optimizer. Candidate portfolios were generated in 5% steps with minimum and maximum weight constraints. Gold was excluded from this optimization universe to test the equity/growth sleeve separately, while Amazon was fixed at 5%. For each candidate portfolio, the code computed expected monthly return, covariance-based volatility, Sharpe, Markowitz utility, max drawdown, HHI concentration and macro betas.



Score component	Weight	Direction
Markowitz utility	35%	higher is better
Markowitz Sharpe	15%	higher is better
Fed beta penalty	20%	lower abs. beta
Inflation beta penalty	15%	lower abs. beta
HHI concentration	10%	lower concentration
Max drawdown	5%	lower loss
Risk aversion lambda	3.0	variance penalty

Selected scenario metrics

Mean monthly return 0.8478%	Monthly volatility 4.2799%	Max drawdown -22.69%	Beta to Fed -0.0280	Beta to inflation -0.2387	Final score 0.7776
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Additional selected metrics: cumulative return 78.73%, macro R-squared 0.4918 and HHI concentration 0.2550.

What a reader should understand

- 1 The project starts from a real portfolio and turns it into a structured risk-diagnostic case study, not a simple performance report.
- 2 The portfolio had a strong equity/growth tilt. Several assets looked different by name, but behaved similarly under correlation analysis.
- 3 Individual VaR/CVaR showed that Amazon and Robotics carried the largest standalone monthly tail risk, while Gold had the lowest monthly CVaR among the assets.
- 4 The macro regression showed that VIX shocks and 10Y Treasury yield changes were the most relevant explanatory factors for portfolio returns in the sample.
- 5 The Markowitz + macro extension used a transparent scoring system that balanced return, volatility, Sharpe, drawdown, concentration and macro sensitivity.

How this project can be positioned in applications

This project is best positioned as a portfolio analytics and risk management case study. It demonstrates Python-based data handling, investment reasoning, downside-risk measurement, macro-factor modeling and constrained allocation design. It is particularly relevant for asset management, wealth management, risk management, fintech analytics and investment research roles.

Limitations

The analysis is backward looking and based on the available historical sample. The Monte Carlo simulation is distributional not predictive. The regression is only descriptive. The Markowitz analysis is a

One-line CV description

Built a Python portfolio analytics framework using FactSet data to assess allocation quality, tail risk, macro sensitivity and constrained Markowitz-style allocation improvements.